FSAR 2026 - Group Project

I. Overview

Aurora Market is a multi-seller e-commerce platform. Your team will work as an analytics group using the company's operational database.

Your task is to:

- become familiar with the data;
- identify a meaningful and feasible research question;
- prepare and analyse the relevant data in R;
- communicate your findings clearly;
- explain and defend the analytical decisions made by your group.

The database contains seven related tables. A README and data dictionary are provided with the data files.

Project Schedule:

Date	Stage	Deliverable / Activity
Oct 8	**Project Scoping Package**	2-page Scoping Brief + Initial Data Decision Log + R code
Oct 26–30	**Project Review Meeting with Prof.**	Mandatory group meeting
Nov 13	**Project Status Update**	3 slides + current R code
Dec 1	**Final Project Package**	Analytics Brief + R project + Process Appendix + presentation deck
Dec 4	**Presentation & Defense**	10-minute presentation + discussion

Regular office hours are available throughout the semester.

Grading:

Component	Weight
Project Process and Scoping	**15%**
Final Analytics Brief	**40%**
Reproducibility & Process Documentation	**15%**
Final Presentation & Discussion	**30%**
Total	**100%**

Marks are based on the quality of the analytical process and reasoning. You are **not** expected to obtain statistically significant or commercially convenient results.

Use of AI

AI tools may be used during the project, but your group remains responsible for all submitted code, analysis, interpretation, and written material.

You must understand and be able to explain work produced with AI assistance. **Substantive uses of AI, and how they were verified, must be documented in the final submission.**

During the final presentation and defense, students may be asked to explain **any** part of the submitted work.

II. Submission #1 (October 8): Project Scoping Package

The purpose of this submission is to show that your group has explored the database sufficiently to propose a viable analytical project. It is **not** a complete data-cleaning exercise and it is **not** an early version of the final report. Groups should submit:

1. Scoping Brief (max. 2 pages)

1.1 Data Familiarization: Briefly summarize the most important things your group learned from exploring all seven tables. Your exploration should consider, where relevant:

- table dimensions and possible units of observation;
- identifiers and relationships between tables;
- missing values;
- duplicate or potentially duplicated records;
- unusual numerical or categorical values;
- other issues that could affect later analysis.

Select and explain the findings that matter. You are not expected to identify every possible issue, and an unusual observation is not automatically an error.

1.2 Proposed Project Scope:

- your proposed research question;
- why it matters from a business perspective;
- your proposed unit of observation;
- likely outcome and explanatory variables;
- tables you expect to use;
- how those tables will need to be connected;
- your intended analytical/statistical approach;
- at least one important limitation, data issue, or analytical risk.

The research question may still be refined after instructor feedback.

2. Initial Data Decision Log

Submit up to **10 entries**. For each issue record:

Field	Description
Issue	What you noticed
Evidence	How you investigated it
Possible responses	Reasonable alternatives
Current status	Current decision or `Unresolved`
Possible consequence	Why it may matter
AI involvement / verification	Whether AI contributed and how its suggestion was checked

You do not need to resolve every issue at this stage.

3. R Code

Submit the R code used to produce the evidence discussed in your Scoping Brief. It does not need to contain the final analytical dataset or final statistical models.

III. Project Review Meeting (October 26–30)

Each group will meet with the instructor for approximately **15–20 minutes** during the week of October 26. The meeting will focus primarily on the proposed research question, the feasibility of the planned analysis, and any issues that may affect the group's progress. The purpose is to confirm or refine the project scope before the main analytical stage.

The meeting may also be used to discuss relevant feedback on the Scoping Package, clarify questions about the project, and check how the group is working together. Groups should be prepared to briefly discuss the division of work, member participation, and any difficulties that have arisen. The meeting is formative and is intended to help identify potential problems early rather than to provide a detailed review of the entire project.

IV. Submission #2 (November 13): Project Status Update

Submit a **three-slide management-style update** and your current R code.

Slide 1 — Scope and Status

- Include your current research question, analytical unit/sample, main variables, and any important changes since the Project Review Meeting.

Slide 2 — Evidence So Far

- Present **one meaningful analytical result**: for example, a figure, table, statistical comparison, or model result.
- Briefly explain what it shows and how it relates to your research question.

Slide 3 — Risks and Next Actions

- Identify an important remaining data, statistical, or interpretive issue and the main work that remains before the final submission.
- If no major problem remains, identify what you still need to verify before finalizing the analysis.
- The Status Update is not a draft final report.

V. Submission #3 (December 1): Final Project Package

Submit four components. The **Analytics Brief** is graded primarily under the Final Analytics Brief component, while the **Reproducible R Project** and **Project Process Appendix** together form the Reproducibility & Process Documentation component.

1. Analytics Brief (max. 6 pages)

This is the main written analytical product. It should be written as an **analytics brief for an informed business audience**, not as a traditional academic paper.

The brief should move directly from the research question to the evidence needed to answer it. You do **not** need a cover page, abstract, literature review, theoretical-background section, or lengthy general introduction.

The six-page limit includes all text, tables, and figures. Use this space selectively: include only descriptive statistics, graphs, regression results, and other evidence that contribute meaningfully to understanding the data or answering the research question. Tables and figures should support the written argument rather than replace it.

The brief should communicate:

- **Research Question and Motivation:** What did you investigate? Why is the question relevant for Aurora Market?
- **Data and Analytical Design:** What is the unit of observation? Which tables and variables were used? How was the analytical sample constructed? Describe the main choices that define the final dataset used in the analysis.
- **Descriptive Evidence:** Provide relevant descriptive statistics and visualizations that help the reader understand the final cleaned analytical sample, the key variables, and any important patterns in the data.
- **Statistical Approach:** What methods did you use, and why are they appropriate for the research question and the structure of the data?
- **Results and Interpretation:** Present the main analytical evidence using appropriate graphs, tables, and regression results. The text should explain the important messages from every table or figure used in the brief. Key regression coefficients should be interpreted substantively, including their sign, magnitude, significance, and practical meaning in the context of the research question.
- **Limitations, Robustness, and Uncertainty:** What important limitations, uncertainties, assumptions, or analytical choices affect how the results should be interpreted? Where relevant, discuss whether alternative reasonable choices change the main conclusions.
- **Business Implications and Conclusion:** What can Aurora Market reasonably learn from the analysis? What conclusions are supported by the evidence, and what should the company avoid concluding?

2. Reproducible R Project

Submit the code required to reproduce the substantive analysis presented in the Analytics Brief.

The R project should:

- **Start from the supplied Aurora Market data files.**
- **Reproduce the key data-processing steps**, including relevant joins, cleaning, variable construction, filtering, and the final analytical sample.
- **Reproduce the main outputs** in the Analytics Brief, including descriptive statistics, tables, figures, and statistical models.
- **Use relative file paths.**
- **Include a short README** with the script order and required packages.
- **Be clear enough to follow and explain.** Stylistic perfection is not required, but you are responsible for understanding what the code does.

3. Process Documentation Appendix (max. 4 pages)

The appendix contains:

- **1. Final Data Decision Log**
  - Update the log started during the Scoping stage. Include **up to 10 consequential decisions** affecting the final analysis.

- **2. Things We Almost Got Wrong**
  - Identify **2–3 consequential mistakes, misunderstandings, or analytical choices** that your group reconsidered or corrected.
  - Briefly explain what changed and why it matters.
- **3. AI Use and Verification Log**
  - Document substantive uses of AI that contributed to the submitted work.
  - You do not need to submit complete prompt histories.
  - For each substantive use, briefly record: Tool, Purpose, Contribution used, and How it was verified.

4. Presentation Deck

- one title slide;
- up to **8 content slides**;
- up to **3 backup slides** for the discussion.

VI. Presentation and Oral Defense

Each group will have **exactly 10 minutes** to present its project. Presentations will be stopped at the 10-minute mark, so groups should plan and rehearse accordingly.

The presentation will be followed by approximately **10–15 minutes of discussion and questioning**.

The discussion may cover the research motivation, data construction, analytical sample, cleaning and variable-construction decisions, statistical methods, interpretation of results, limitations, robustness, and any AI-supported work.

Every group member is expected to contribute meaningfully to the presentation, the discussion, or both. Questions may be directed to specific students, and all members should be prepared to explain the group's submitted analysis.

The presentation grade reflects the quality of the slides, oral delivery, and the group's ability to explain and discuss its work.

Substantial differences in individual participation, understanding, or ability to discuss the submitted analysis may result in individual adjustments to the final project grade.

VII. Peer Evaluation

At the end of the project, each student will receive an email invitation to complete a **mandatory peer evaluation through TEAMMATES**.

Each student will distribute **100 points across all members of the group, including themselves**, to reflect their assessment of each member's overall contribution to the project. An equal contribution would therefore correspond to an equal division of the 100 points among group members.

Students may provide comments explaining their allocations. Comments are optional when contributions were broadly similar. If you allocate substantially more or fewer points to a group member than an equal share, you should provide a brief justification.

Peer evaluations are used as evidence of individual contribution. **Consistent evidence that members contributed substantially differently may result in individual adjustments to the final project grade.** Adjustments will not normally be based on a single rating in isolation; peer evaluations may be considered together with other relevant evidence from the project process, group meetings, presentation, and discussion.